

HALLUCINATION MEASUREMENT INDEX FRAMEWORK FOR BILINGUAL

MALAY-ENGLISH MENTAL HEALTH CHATBOTS

ABSTRACT

Large language models (LLMs) powering AI chatbots are increasingly deployed for mental health support, yet they remain prone to generating hallucinated content. Responses that are factually incorrect, clinically ungrounded, or contextually inappropriate present a risk to users seeking mental health support. This study proposes the Hallucination Measurement Index (HMI), a composite scoring index designed to measure the level of hallucination in AI chatbot responses within the mental health domain. The HMI is a student-designed composite index, representing the original contribution of this research. The HMI integrates five automated sub-metrics: Factual Consistency, Response Groundedness, Safety Compliance, Semantic Coherence, and Cultural Appropriateness, each operationalised using established NLP tools adapted for the Malaysian bilingual context. A custom bilingual scenario bank (MH-Hallu-MY) comprising 305 mental health prompts across 12 clinical categories is administered to three widely accessible LLM platforms powered by GPT-4 (OpenAI), Gemini Pro (Google DeepMind), and GPT-4 Turbo (Microsoft), generating 2,745 responses to evaluate and test the HMI index. The study employs non-parametric statistical tests including the Kruskal-Wallis H-test and Dunn's post-hoc analysis with Bonferroni correction to compare hallucination profiles across platforms. A Streamlit-based evaluation prototype operationalises the HMI for reproducible assessment. This research addresses a critical gap: the absence of a domain-specific, composite hallucination measurement index for mental health AI in Malaysia.

Keywords: Hallucination Measurement, AI Chatbots, Mental Health, Large Language Models, Malaysia.